



## counting sort 12

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Complete the countingSort function.

countingSort has the following parameter:

int arr[n]: an array of integers Returns Nothing. Print the sorted array with each element separated by a space.

### Input Format

The first line contains an integer  $n$ , the number of elements in the array. The second line contains  $n$  space-separated integers. 10 11 3 2 1 4 5 6 4 3

### Constraints

$1 \leq n \leq 100000$   $0 \leq arr[i] < 100$

### Output Format

1 1 1 2 3 3 4 4 5 6

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Language

C



```
1  #include <stdio.h>
2  int main() {
3      int n;
4      scanf("%d", &n);
5      int arr[n];
6      for (int i = 0; i < n; i++) {
7          scanf("%d", &arr[i]);
8      }
9      int freq[100] = {0};
10     for (int i = 0; i < n; i++) {
11         freq[arr[i]]++;
12     }
13     for (int i = 1; i < 100; i++) {
14         freq[i] += freq[i - 1];
15     }
16     int res[n];
17     for (int i = n - 1; i >= 0; i--) {
18         res[freq[arr[i]] - 1] = arr[i];
19         freq[arr[i]]--;
20     }
21     for (int i = 0; i < n; i++) {
22         printf("%d ", res[i]);
23     }
24     return 0;
25 }
```

Line: 25 Col: 1

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